

Design and Simulation of a Silicon Photonic Mach–Zehnder Interferometer

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Abstract

This preliminary report summarizes the design and simulation of a silicon photonic Mach–Zehnder interferometer (MZI). Finite-difference eigenmode (FDE) simulations are used to characterize the quasi-TE and quasi-TM waveguide modes. A wavelength-dependent compact model is fitted to the effective index, and the effects of waveguide width and thickness variations are evaluated. The resulting MZI transmission is calculated in MATLAB and compared with circuit-level simulations in Lumerical INTERCONNECT. Several arm-length differences are considered for the proposed design.

1 Introduction

The spectral response of a Mach–Zehnder interferometer depends on the phase difference accumulated between its two arms. Fabrication-induced variations in waveguide width and thickness affect the effective index and, consequently, the positions and spacing of the transmission fringes. This work examines these effects for a nominal silicon SOI waveguide of width 500 nm and thickness 220 nm.

The study begins with an FDE modal analysis. The extracted waveguide properties are then used to calculate the MZI response and to assess the impact of geometrical variability. Finally, the analytical response is compared with results from Lumerical INTERCONNECT.

2 Waveguide Modal Analysis

2.1 Mode profiles

Figure 1 shows the field-amplitude profiles of the fundamental quasi-TE and quasi-TM modes at the central wavelength $\lambda_0 = 1.55 \mu\text{m}$. The spatial refractive-index profile of the nominal waveguide is also shown.

2.2 Effective index and dispersion

The effective index is calculated over the wavelength range 1.50 μm to 1.60 μm . Around λ_0 , a quadratic approximation is written as

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2, \quad (1)$$

where λ and λ_0 are expressed in μm . Therefore, n_2 has units of $1/\mu\text{m}$ and n_3 has units of $1/\mu\text{m}^2$.

The group index follows from

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \quad (2)$$

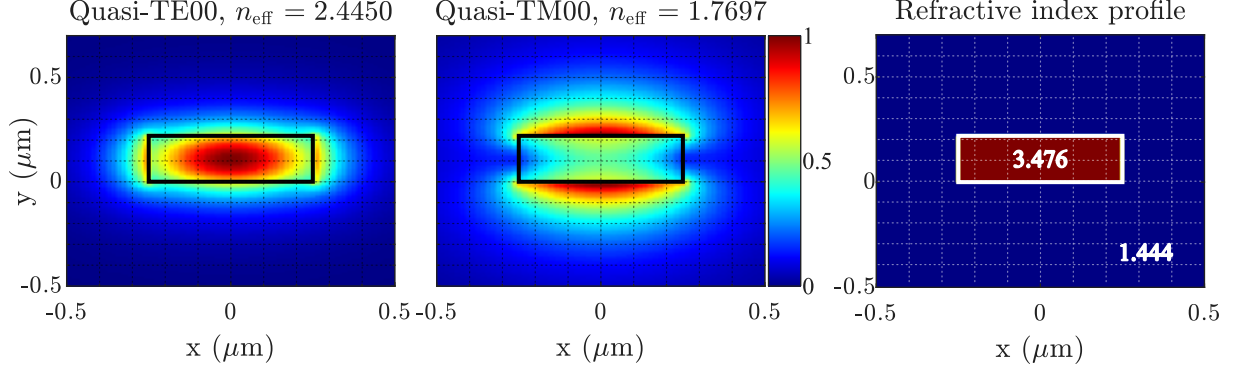


Figure 1: Field-amplitude profiles of the fundamental quasi-TE and quasi-TM modes and refractive-index profile of the nominal 500 nm by 220 nm waveguide at $\lambda_0 = 1.55 \mu\text{m}$.

At the expansion wavelength,

$$n_g(\lambda_0) = n_1 - \lambda_0 n_2. \quad (3)$$

The chromatic dispersion is

$$D(\lambda_0) = -\frac{\lambda_0}{c} \frac{d^2 n_{\text{eff}}}{d\lambda^2} \Big|_{\lambda_0}. \quad (4)$$

The wavelength and polynomial coefficients must be converted to SI units when evaluating Eq. (4) in s/m^2 .

Table 1: Quadratic-fit coefficients of the effective index around $\lambda_0 = 1.55 \mu\text{m}$. Wavelength is expressed in μm in Eq. (1).

Mode	n_1	n_2 ($1/\mu\text{m}$)	n_3 ($1/\mu\text{m}^2$)
Quasi-TE	2.4449	-1.1305	-4.2067
Quasi-TM	1.7697	-1.2687	1.8448

Figure 2 presents the simulated effective and group indices. The effective index is plotted against the left vertical axis, and the group index against the right vertical axis. The quasi-TE mode has a higher effective index than the quasi-TM mode throughout the investigated wavelength range. In the fitted model, the quasi-TE group index increases with wavelength, whereas the quasi-TM group index decreases.

3 Waveguide Geometry Variability

A corner analysis is performed around the nominal waveguide geometry using nine combinations of width and thickness. Figure 3 shows the corresponding wavelength-dependent effective and group indices.

Table 2 lists the minimum and maximum values among the nine geometries at $\lambda_0 = 1.55 \mu\text{m}$. The effective-index values are obtained from the fitted model, whereas the reported group-index extrema are obtained from dedicated FDE calculations at this wavelength.

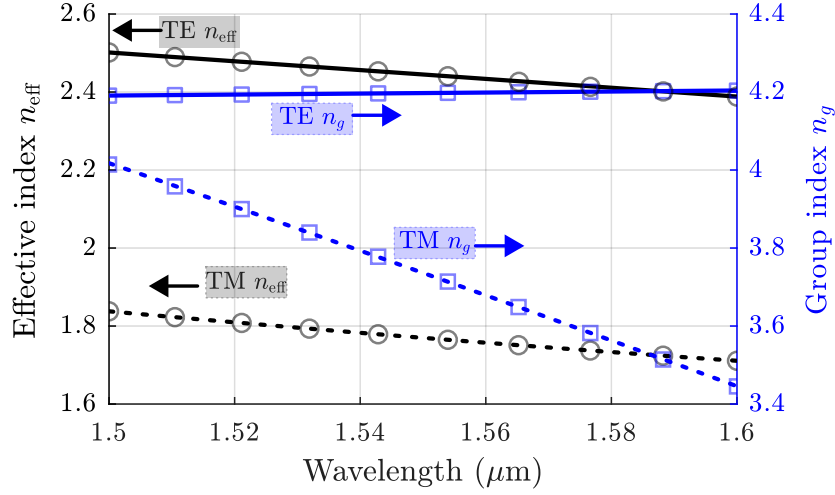


Figure 2: Effective index and group index of the quasi-TE and quasi-TM modes over the wavelength range 1.50 μm to 1.60 μm .

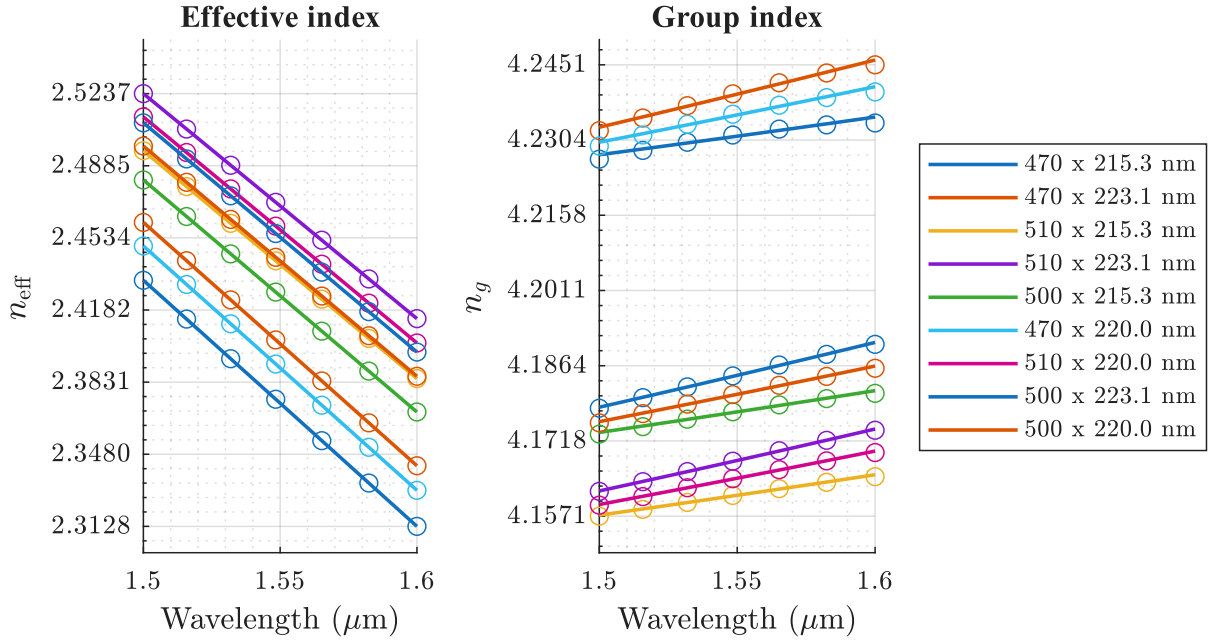


Figure 3: Effective and group indices for the nine investigated waveguide geometries. The curves show the fitted model and the markers represent FDE results.

Table 2: Index extrema among the nine geometries at $\lambda_0 = 1.55 \mu\text{m}$.

Quantity	Value	Width (nm)	Thickness (nm)
Minimum n_{eff} (fit)	2.37282985	470	215.3
Maximum n_{eff} (fit)	2.46894739	510	223.1
Minimum n_g (FDE)	4.16128925	510	215.3
Maximum n_g (FDE)	4.23964575	470	223.1

4 Mach–Zehnder Interferometer

4.1 Transmission model

Let L_1 and L_2 denote the lengths of the two MZI arms. Assuming ideal 50:50 couplers and a power-attenuation coefficient α , the normalized power transmission at the considered output is

$$T(\lambda) = \frac{1}{4} \left| e^{-\alpha L_1/2} e^{-i\beta(\lambda)L_1} + e^{-\alpha L_2/2} e^{-i\beta(\lambda)L_2} \right|^2, \quad (5)$$

where

$$\beta(\lambda) = \frac{2\pi n_{\text{eff}}(\lambda)}{\lambda}. \quad (6)$$

The transmission plotted in decibels is

$$T_{\text{dB}}(\lambda) = 10 \log_{10}(T(\lambda)). \quad (7)$$

For an arm-length difference $\Delta L = |L_2 - L_1|$, the local free spectral range (FSR) near λ_0 is approximately

$$\Delta\lambda_{\text{FSR}} \simeq \frac{\lambda_0^2}{n_g(\lambda_0)\Delta L}, \quad \Delta f_{\text{FSR}} \simeq \frac{c}{n_g(\lambda_0)\Delta L}. \quad (8)$$

4.2 Variability and circuit-level comparison

Figure 4 shows the calculated quasi-TE MZI transmission for $\Delta L = 50 \mu\text{m}$. The nominal geometry is highlighted in blue, and the other eight geometries in gray illustrate the effect of fabrication variability. Variations in the effective index shift the spectral positions of the interference fringes, while variations in group index modify their spacing.

The model is also compared with two Lumerical INTERCONNECT simulations. The dashed red curve excludes the grating-coupler input/output response. The solid black curve includes that response, with a reported insertion-loss contribution of approximately -5.18 dB and a wavelength-dependent spectral envelope. The two INTERCONNECT spectra exhibit approximately the same FSR. A small shift in fringe position relative to the MATLAB model is observed.

Table 3 summarizes the calculated local FSR at $\lambda_0 = 1.55 \mu\text{m}$ for the nine geometries and $\Delta L = 50 \mu\text{m}$.

Table 3: Effective index, group index, and calculated local FSR at $1.55 \mu\text{m}$ for the nine waveguide geometries. The arm-length difference is $50 \mu\text{m}$.

Case	Width (nm)	Thickness (nm)	n_{eff}	n_g	FSR (nm) / FSR (GHz)
1	470	215.3	2.3728	4.2312	11.356 / 1417.1
2	470	223.1	2.4017	4.2394	11.334 / 1414.3
3	510	215.3	2.4404	4.1612	11.547 / 1440.9
4	510	223.1	2.4689	4.1680	11.528 / 1438.6
5	500	215.3	2.4251	4.1774	11.502 / 1435.3
6	470	220.0	2.3900	4.2353	11.345 / 1415.7
7	510	220.0	2.4573	4.1645	11.538 / 1439.7
8	500	223.1	2.4537	4.1846	11.483 / 1432.8
9	500	220.0	2.4421	4.1809	11.493 / 1434.1

5 Conclusion and Outlook

The FDE analysis and fitted waveguide model provide the effective index and group index needed to predict the MZI response. The geometry study shows that fabrication variations affect both

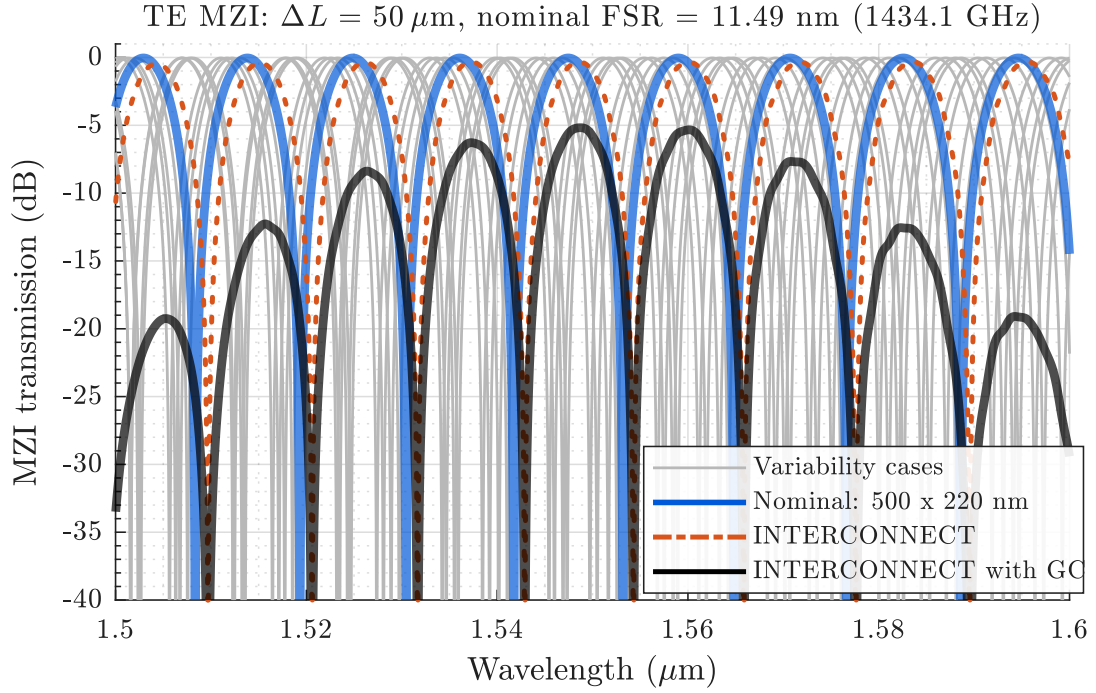


Figure 4: Quasi-TE MZI transmission for an arm-length difference of $50 \mu\text{m}$. The MATLAB nominal response and geometry variations are compared with two INTERCONNECT simulations, respectively excluding and including the grating-coupler response.

the fringe positions and the local FSR. Circuit-level INTERCONNECT simulations provide an additional comparison and allow the influence of the grating-coupler response to be examined.

The proposed design study will consider quasi-TE MZIs with arm-length differences of $10 \mu\text{m}$, $20 \mu\text{m}$, $50 \mu\text{m}$, $100 \mu\text{m}$, and $200 \mu\text{m}$. Further work may include a 2-by-2 multimode-interference coupler and an assessment of a quasi-TM implementation.